The **Future** Is **Paper**

The future of displays could see a radical shift in technology. Coming soon: displays as thin as a human hair!

Robert Sovereign-Smith

In *Gadgetopia* (*Digit*, July 2005), we designed the theoretical Gadget of Our Dreams (GOD)—basically, a single gadget most of us would like to have, provided someone put it together and marketed it. It was conceptualised using existing technologies, one of which was the paper/foldable display.

Thus far, gadget sizes have been defined by the size of their screens. Too large a screen and the gadget cannot be sold as a 'portable' device; too small, and the screen becomes barely readable, ruling out full-fledged computing on it! This, perhaps, is why there's still a market for PDAs, despite the fact that mobile phones—which have a much more compact form factor—have enough processing power to accomplish the tasks of a regular PDA.

We may not yet be at the stage of bringing high-end 3D gaming to a mobile phone, but we certainly have the technology and hardware capabilities to make devices powerful yet small. Again, the problem has always been displays: a display that's too large not only results in an unacceptable form factor, but also eats up battery power by the watts.

But what if there was a technology that could provide you with a large display screen which could be rolled, folded and manipulated to fit into a tiny space when not being used? And at the same time it used less battery power than the backlight of your cell phone?

Amazing, right? Well, not quite: this is old news now. The problem is that the black-and-white era is behind us, and we're not interested in going back there—most such displays are only monochrome capable yet, with the operative word being "yet"!

The Legacy

The idea of paper displays, or ePaper as it's also known, is hardly new. Back in 1975, Nicholas Sheridan, a physicist working at the Xerox PARC (Palo Alto Research Center), started his

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